

CONCEPT CHECK 18.4

1. **MAKE CONNECTIONS** As you learned in Chapter 12, mitosis gives rise to two daughter cells that are genetically identical to the parent cell. Yet you, the product of many mitotic divisions, are not composed of identical, zygote-like cells. Why?
2. **MAKE CONNECTIONS** Explain how the signaling molecules released by an embryonic cell can induce changes in a neighboring cell without entering the cell. (See Figures 11.15 and 11.16.)
3. How do fruit fly maternal effect genes determine the polarity of the egg and the embryo?
4. **WHAT IF?** In Figure 18.17b, the lower cell is synthesizing signaling molecules, whereas the upper cell is expressing receptors for these molecules. In terms of gene regulation and cytoplasmic determinants, explain how these cells came to synthesize different molecules.

For suggested answers, see Appendix A.

CONCEPT 18.5

Cancer results from genetic changes that affect cell cycle control

In Concept 12.3, we considered cancer as a disease in which cells escape from their normal cell division control mechanisms. Now that we have discussed the molecular basis of gene expression and its regulation, we can look at cancer more closely. The gene regulation systems that go wrong during cancer turn out to be the very same systems that play important roles in embryonic development, the immune response, and many other biological processes. Thus, research into the molecular basis of cancer has both benefited from and informed many other fields of biology.

Types of Genes Associated with Cancer

The genes that normally regulate cell growth and division during the cell cycle include genes for growth factors, their receptors, and the intracellular molecules of signaling pathways. (To review cell signaling, see Concept 11.2; for regulation of the cell cycle, see Concept 12.3.) Mutations that alter any of these genes in somatic cells can lead to cancer. The agent of such change can be random spontaneous mutation. Many cancer-causing mutations likely also result from environmental influences: chemical carcinogens like tobacco, X-rays and other high-energy radiation, and some viruses.

Cancer research uncovered cancer-causing genes called **oncogenes** (from the Greek *onco*, tumor) in certain types of viruses. Later, related versions of viral oncogenes were found in the genomes of humans and other animals. The normal versions of the cellular genes, called **proto-oncogenes**, code for proteins that stimulate normal cell growth and division.

How might a proto-oncogene—a gene that has an essential function in normal cells—become an oncogene, a cancer-

causing gene? In general, an oncogene arises from a genetic change that leads to an increase either in the amount of the proto-oncogene's protein product or in the intrinsic activity of each protein molecule. The genetic changes that convert proto-oncogenes to oncogenes fall into four main categories (**Figure 18.23**): epigenetic changes, translocations, gene amplification, and point mutations.

First, alterations in epigenetic modifications that can lead to abnormal chromatin condensation in a cell are often found in tumor cells. If a mutation in a gene for a chromatin-modifying enzyme leads to loosening of chromatin in a region that is normally not being expressed, a proto-oncogene in that region could be expressed at abnormally high levels (see Figure 18.23a). For example, a gene for one such enzyme has been shown to be mutated in 20% of tumor cells analyzed.

Second, cancer cells are frequently found to contain chromosomes that have broken and rejoined incorrectly, translocating fragments from one chromosome to another (see Figure 15.14). If a translocated proto-oncogene ends up near an especially active promoter (or other control element), its transcription may increase, making it an oncogene (see Figure 18.23b).

The third main type of genetic change, amplification, increases the number of copies of the proto-oncogene in the cell through repeated gene duplication (discussed in Concept 21.5; see Figure 18.23c).

Fourth, a point mutation either in the promoter or an enhancer that controls a proto-oncogene, could cause an increase in its expression. A point mutation in the coding sequence of the proto-oncogene could change the gene's product to a protein that is more active or more resistant to degradation than the normal protein (see Figure 18.23d).

Any of these four mechanisms can lead to abnormal stimulation of the cell cycle and put the cell on the path to becoming a cancer cell.

In addition to genes whose products normally promote cell division, cells contain genes whose normal products *inhibit* cell division. Such genes are called **tumor-suppressor genes** since the proteins they encode help prevent uncontrolled cell growth. Any mutation that decreases the normal activity of a tumor-suppressor protein may contribute to the onset of cancer, in effect stimulating growth through the absence of suppression.

The protein products of tumor-suppressor genes have various functions. Some repair damaged DNA, which prevents the cell from accumulating cancer-causing mutations. Other tumor-suppressor proteins control adhesion of cells to each other or to the extracellular matrix; proper cell anchorage is crucial in normal tissues and is often absent in cancers. Still other tumor-suppressor proteins are components of cell-signaling pathways that inhibit the cell cycle.